

FL Simulation Results

File: test_B3

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 6
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 34.0% (0.34)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 34.0% (0.34) of data for 15 rounds with 6 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8751
Clean + DP Protection Accuracy: 0.8741
Poisoned Accuracy: 0.8395
Poisoned + DP Protection Accuracy: 0.8402
Drop (Clean - Poisoned): 0.0356
Drop (Clean - Init): 0.0823
Drop (Clean DP - Init): 0.0813
Drop (Poisoned - Init): 0.0467
Drop (Poisoned DP - Init): 0.0474
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8804
Clean Recall: 0.8751
Clean F1 Score: 0.8745

Clean DP Precision: 0.8799
Clean DP Recall: 0.8741
Clean DP F1 Score: 0.8735

Poisoned Precision: 0.8480
Poisoned Recall: 0.8395
Poisoned F1 Score: 0.8385

DP Protection Precision: 0.8487
DP Protection Recall: 0.8402
DP Protection F1 Score: 0.8394

Summary

FL Simulation Complete
Task: 3052550b-67bf-47cc-9aa0-adbd8ee666a9
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8751
Clean DP Accuracy: 0.8741
Poisoned Accuracy: 0.8395
Data Poison Protection Accuracy: 0.8402
Drop (Clean - Poisoned): 0.0356
Drop (Clean - Init): 0.0823
Drop (Clean DP - Init): 0.0813
Drop (Poisoned - Init): 0.0467
Drop (Poisoned_DP - Init): 0.0474
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8804
Clean Recall: 0.8751
Clean F1 Score: 0.8745
Clean DP Precision: 0.8799
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Poisoned Precision: 0.8480
Poisoned Recall: 0.8395
Poisoned F1 Score: 0.8385
DP Protection Precision: 0.8487
DP Protection Recall: 0.8402
DP Protection F1 Score: 0.8394